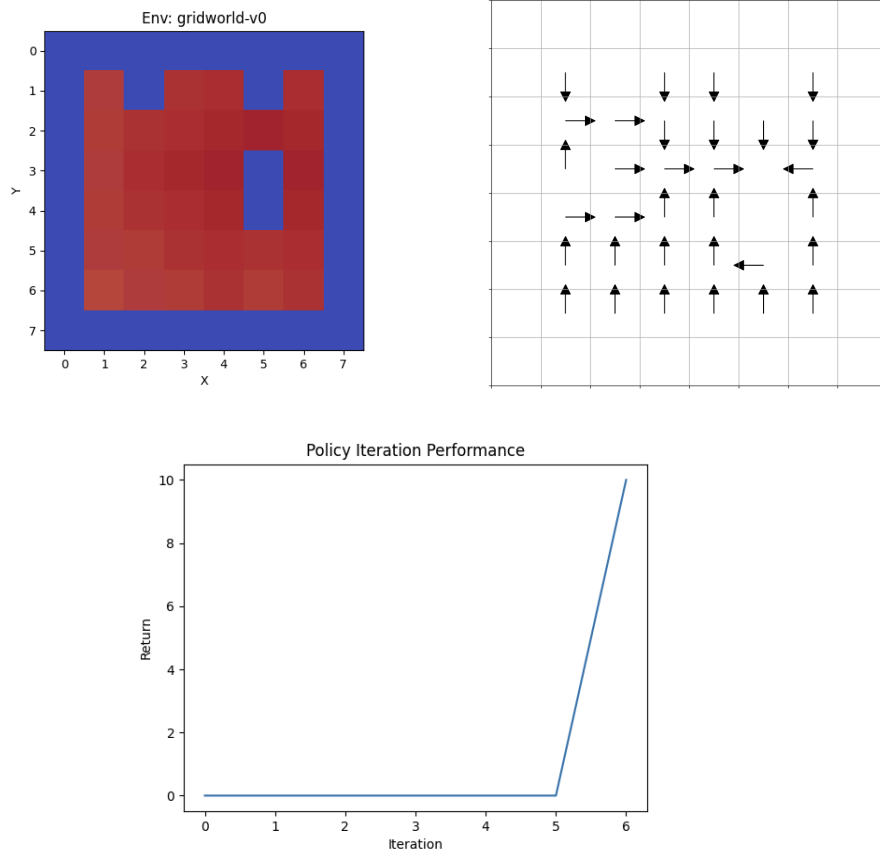


Advanced Robotics - HW3 Writeup

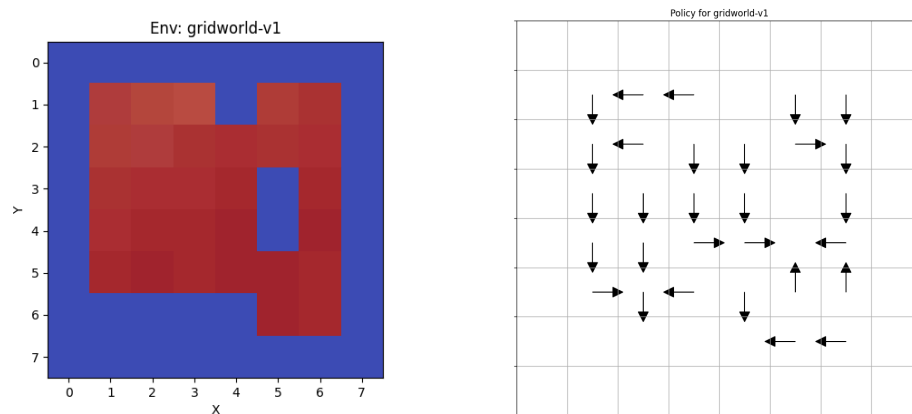
Joseph Neal

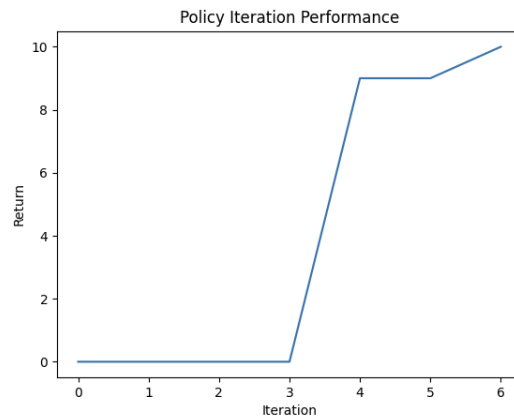
2.1: Policy Iteration for Gridworld Environment

Gridworld 0:



Gridworld 1:



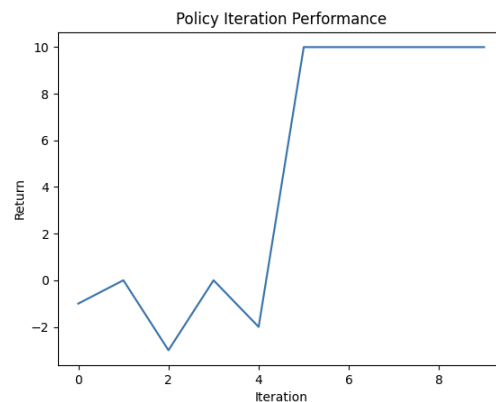
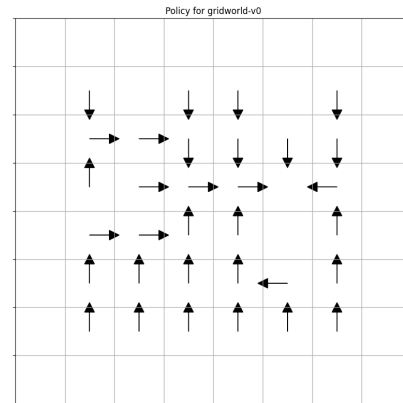
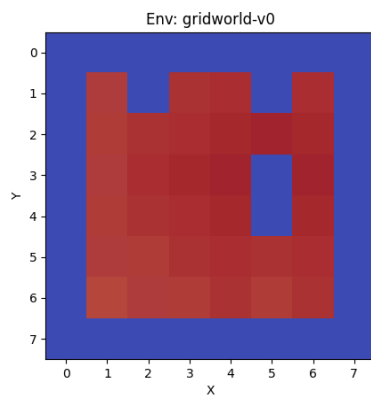


Written Questions:

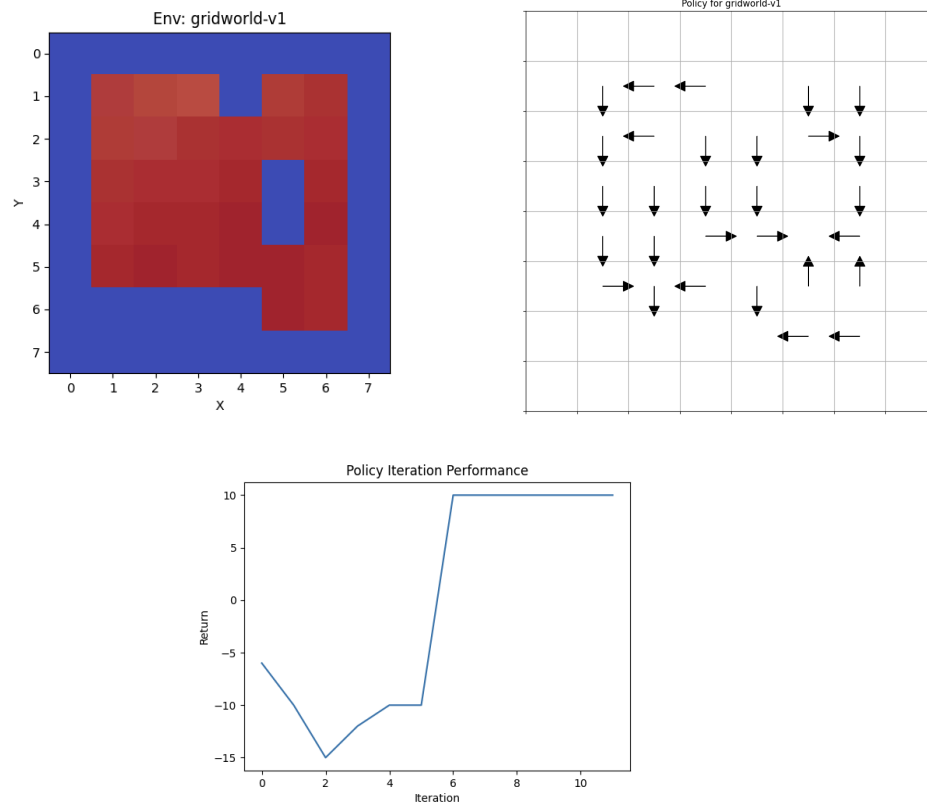
1. Lowering gamma does change the value function, effectively making the heatmap's gradient less steep. Generally, changing the value function does not affect the policy, however if lowered enough, the policy/value function does not converge before max iterations occur, as the agent cannot see far enough into the future.

2.2: Value Iteration for Gridworld Environment

Gridworld 0:



Gridworld 1:

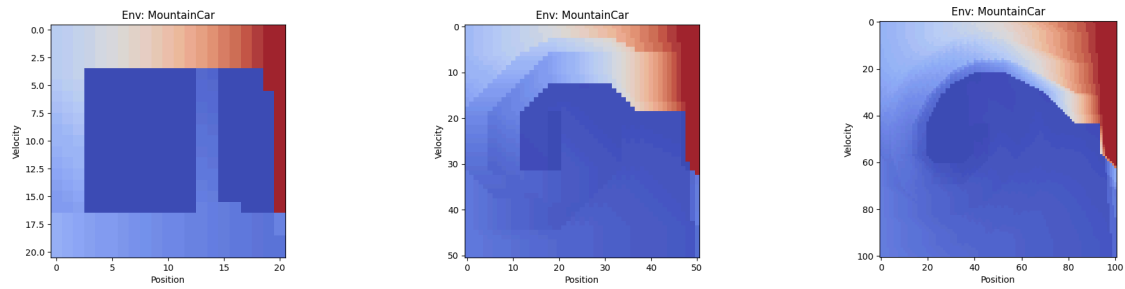


Written Questions:

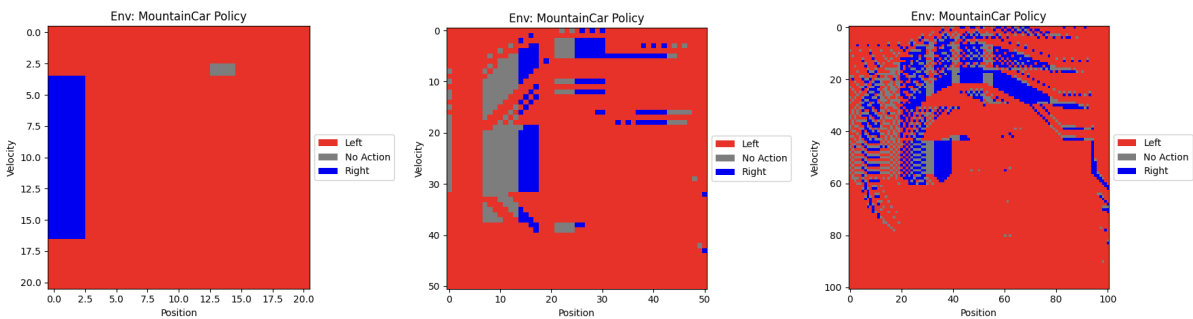
1. Results appear the same for policy and value iteration, with the exception that value iteration converges more quickly. This is to be expected, as both should eventually achieve the most optimal policy and value function.
2. Policy iteration does run slower than value iteration (~140ms vs ~60ms), which is what we would expect as in value iteration, the Optimal Bellman Update immediately incorporates greedy improvements in one step.

3.1: Nearest Neighbor Interpolation

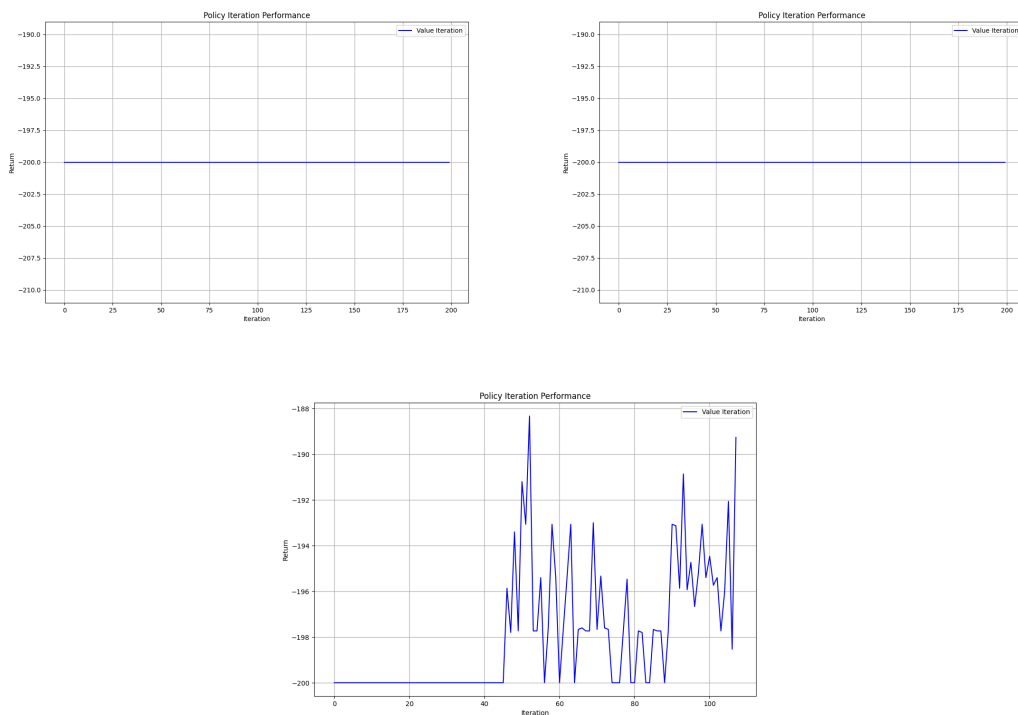
Value function plots for 21 (left), 51 (middle), and 101 (right) bins



Policy plots for 21 (left), 51 (middle), and 101 (right) bins

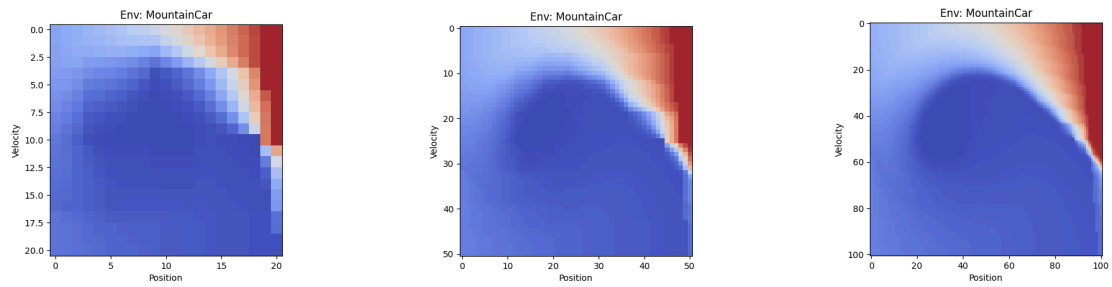


Learning curves for 21 (top left), 51 (top right), and 101 (bottom) bins

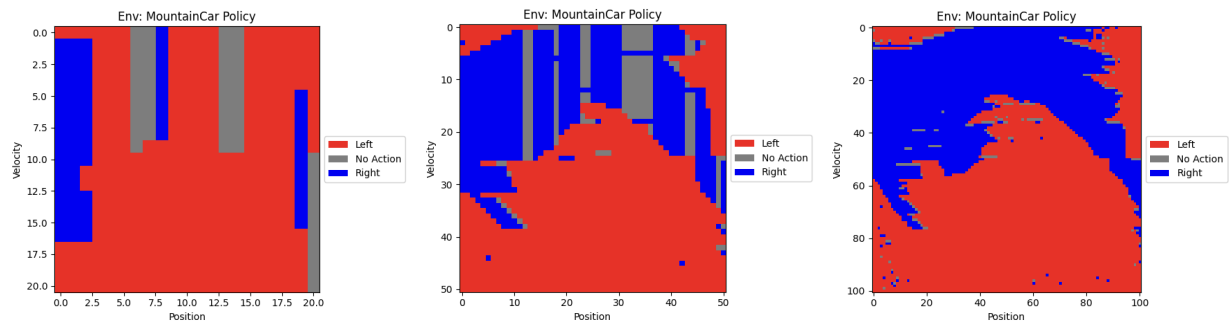


3.2: Linear Interpolation

Value function plots for 21 (left), 51 (middle), and 101 (right) bins



Policy plots for 21 (left), 51 (middle), and 101 (right) bins



Learning curves for 21 (top left), 51 (top right), and 101 (bottom) bins

